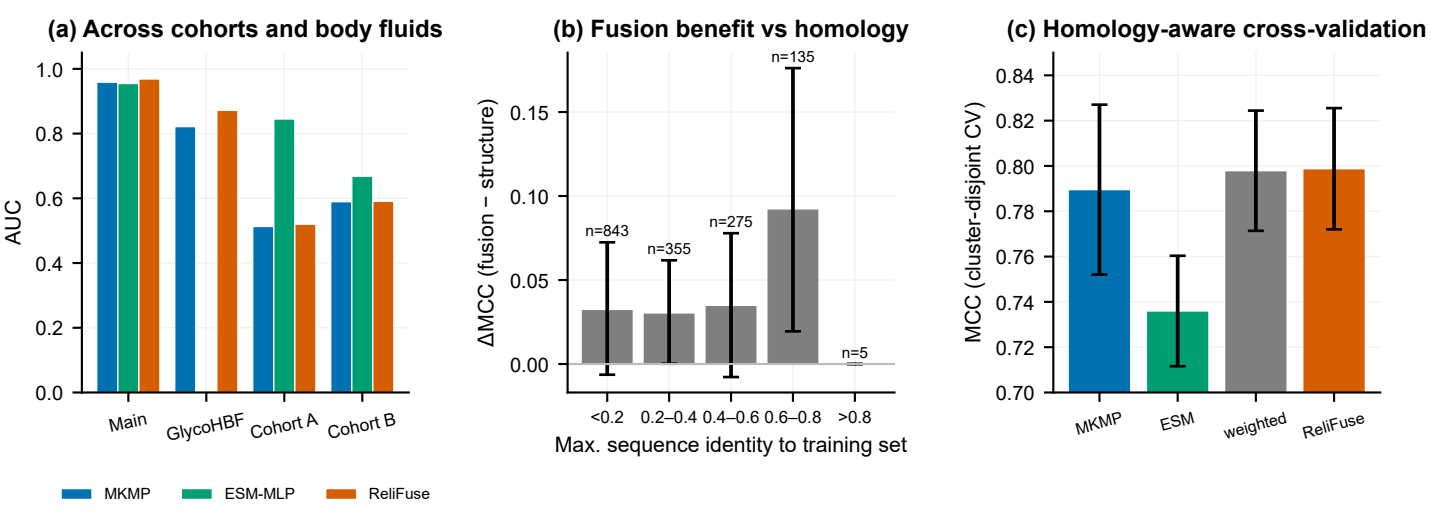
(a) Across cohorts and body fluids
AUC
1.0
0.8
0.6
0.4
0.2
0.0
Main
GlycoHBF
Cohort A
Cohort B
MKMP
ESM-MLP
ReliFuse
(b) Fusion benefit vs homology
ΔMCC (fusion − structure)
0.15
0.10
0.05
0.00
n=843
n=355
n=275
n=135
n=5
<0.2
0.2–0.4
0.4–0.6
0.6–0.8
>0.8
Max. sequence identity to training set
(c) Homology-aware cross-validation
MCC (cluster-disjoint CV)
0.84
0.82
0.80
0.78
0.76
0.74
0.72
0.70
MKMP
ESM
weighted
ReliFuse